

Multiscale Fusion for Spatially Decoupled Multimodal MRI Super-Resolution Reconstruction

Qi-Hao Xu (徐麒皓)^{ID} and Bo Li (李波)^{ID}

Abstract—Super-resolution (SR) is primarily tailored for single-modal medical images. While in many applications of magnetic resonance imaging (MRI), multimodal images with diverse parameters are available. Current single-modal SR approaches are limited in fully exploring and exploiting the correlations between these cross-modal images, leading to degraded reconstruction performance. Thus, this article presents a novel multiscale (MS) fusion approach for cross-modal MRI SR reconstruction. The proposed method develops domain-specific convolutions to spatially decouple MRI into different subspaces and task-specific modules for reconstruction. Specifically, a CNN-based framework is constructed to explore the mapping between a low-resolution T2-weighted (LR T2w) image and a high-resolution (HR) T2w one, by incorporating an HR T1-weighted (T1w) image. In the proposed network, the low-frequency filtering modules have been integrated into it to remove the low-frequency components of the HR T1w while extracting its high-frequency information. Therefore, by fusing the detailed features of the HR T1w and the LR T2w at two different scales, the network generates the HR T2w image. Extensive results across benchmark MRI datasets demonstrate the effectiveness of the proposed method in MRI SR reconstruction.

Index Terms—Convolutional neural network (CNN), magnetic resonance imaging (MRI), multimodal, multiresolution analysis (MRAs), super-resolution (SR).

I. INTRODUCTION

MAGNETIC resonance imaging (MRI) is a noninvasive, radiation-free vision-based measuring instrument, widely used in clinical diagnoses. MRI is renowned for providing detailed contrast images of various soft tissues without exposing patients to ionizing radiation. However, the standard MRI measurement method captures the spatial-frequency information by sampling from the entire k -space, easily leading to low efficiency on image collection. Patients' movement or physiological activities, such as respiration, cardiac motion, and blood flow, often introduce significant artifacts into MRIs [1]. Moreover, the prolonged scanning durations not only increase medical costs but also limit their availability [2]. In addition, MRIs with varied contrasts are essential for making diagnoses of the same region of interest;

however, the low quality of multimodal MRIs often reduces the accuracy. All these necessitate the efficient MRI high-resolution (HR) reconstruction.

Multimodal MRI reconstruction obtains the HR images by removing redundancy in single MRI and leveraging shared information across different MRI sequences [3]. Recent studies [4], [5] have demonstrated that reconstructing under-sampled MR sequences (target modality) with additional information from a fully sampled modality improves the outcomes. These approaches not only simplify MRI process by under-sampling the target in k -space but also enhance overall reconstruction quality.

Above methods have succeeded in multimodal MRI reconstruction; however, they fail to address spatial misalignment between modalities to further improve the final reconstruction performance. Spatial misalignment is very common, especially for dealing with various MRI sequences. Previous studies have attempted to address this problem. For example, Lai et al. [6] employed multimodal image registration to align the reference and the intermediate reconstruction outcome of the undersampled target. But traditional image registration may not be suitable for the undersampled MRIs. Experimental results of multimodal MRI reconstruction have been discussed based on the registered multicontrast MRIs that are free from misalignment [7], [8], while such conditions are not always available in real-world scenarios. Recently, deep neural networks have shown promise in enhancing multimodal MRI reconstruction, while the spatial misalignment issue still needs to be addressed [9], [10]. Therefore, further study on spatial misalignment is required for multimodal MRI reconstruction.

To address above issues, a multiscale fusion spatially decoupled (MFSD) method will be modeled to reconstruct zero-filled MRI in a multimodal setting. MFSD will customize convolutional layers for domain-specific tasks, serving as spatial and angular feature extractors (AFE). Then, the spatially decoupled network computes the displacement between the fully sampled and the decoupled MRIs. The latter, instead of the original misaligned ones, is merged with the undersampled MRIs and then fed into the reconstruction network to generate a high-quality target image. This approach integrates multimodal data to capture a broad range of global information and directs it along specific merging paths based on the characteristics of multimodal discreteness, thereby enhancing the network's representation and specificity. Furthermore, 2-D angular dimension representing structural information and views of MRIs, is also learned by the proposed networks. Multiresolution analysis (MRA) theory [38] posits that those intricate features obtained from the HR T1w and LR T2w

Received 29 November 2024; revised 20 January 2025; accepted 4 February 2025. Date of publication 26 March 2025; date of current version 16 April 2025. The Associate Editor coordinating the review process was Dr. Marco Carratù. (Corresponding author: Bo Li.)

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Digital Object Identifier 10.1109/TIM.2025.3554850

images can be integrated at two distinct scales. This integration facilitates the development of an end-to-end mapping, which effectively correlates the LR T2w images to their HR counterparts, that is, the HR T2w images. In this process, the HR T1w images are also incorporated into them as vital sources of elaborate prior information. This integration supports the generation of the HR T2w images through an upsampling mechanism using dense connectivity blocks, enhancing the overall imaging resolution. To further improve the extraction of high-frequency details, a low-frequency filter module is introduced into the architecture. It is specifically designed to mitigate potential interference stemming from the HR T1w images during capturing the high-frequency information. As a result, the finer details remain, thus facilitating a more accurate synthesis of the HR T2w images.

In sum, this article focuses on developing the proposed MFSD to reconstruct high-quality image from cross-modal MRIs, with comparatively few network parameters and low computational payments. Namely, MFSD is in practical a machine learning-based indirect measurement method, generating the HR T2w images by using the results of vision-based measurement (VBM) [45], that is, the LR T2w and HR T1w images. Therefore, MFSD is very valuable from the viewpoint of instrumentation and measurement [44]. Its main contributions are listed below.

- 1) A module is designed to spatially decouple MRI into different kinds of features, resulting in HR reconstruction by aligning them and the undersampled features.
- 2) A specialized module, termed low-frequency information filtering (LFIF), is developed to eliminate the low-frequency component from the HR T1w, the rest of which is then incorporated into HR T2w image reconstruction.
- 3) A deep network is constructed to fuse features at two scales, and then to generate the HR T2w as result of VBM.

II. RELATED WORK

A. MRI Reconstruction

Image reconstruction aims to generate high-quality images from incomplete k -space MR signals. Conventional linear filtering and zero-filling methods assist in recovering the missing signals; however, they may introduce artifacts into the final images [11], [12]. Compressed sensing (CS) has revolutionized undersampled reconstruction, enabling faster MRI. Recently, there has been a notable increase in deep learning frameworks for MRI reconstruction. Initially, Wang et al. [13] trained a convolutional neural network (CNN) to offline convert zero-filled, undersampled MRIs into fully sampled images. To improve the performance, they incorporated additional modules, such as data consistency layers [14] and recurrent neural networks (RNNs) [15]. Researches also adopted the image space, widely used in CS-MRI, for reconstruction. Han et al. [16] estimated the missing MRI signals in the k -space domain, where MRI can be initially acquired. Zhang et al. [17] processed MRI signals in both the image space and the k -space. A deep learning method predicted that the fully sampled MRIs can be partially acquired from the k -space data, instead of using Fourier transform [18]. The

recent NYU fast MRI, supported by extensive public datasets, has significantly accelerated the progress [19], [20], [21], [22].

Aforementioned methods just reconstructed MRIs from single-modal images; however, much more information hidden in multiply modalities cannot be explored and exploited.

B. Reference-Based MRI Reconstruction

The potential to enhance undersampled MRI with additional information from reference images is promising in MRI reconstruction. Souza et al. [23] demonstrated the accelerated MRI for repeated visits on the same patient by early capturing intrasubject images. Weizman et al. [24] introduced a more flexible approach applied to scenarios where reference data were not fully available. Schlemper et al. [14] explored temporal correlations by leveraging redundancies among consecutive temporal frames of dynamic MRI via 3-D convolution. In contrast, Hirabayashi et al. [25] utilized the redundancy between neighboring 2-D slices in a 3-D volume for reconstruction, recommending comprehensive sampling from the slices rather than undersampling others.

It can improve the reconstruction performance of a specific MRI by utilizing information from other modalities. Majumdar and Ward [26] and Huang et al. [27] reconstructed the T1w and T2w images from undersampled MRIs in the same area. Song et al. [28] suggested joint dictionary learning to reconstruct multicontrast MRI images. Recently Xiang et al. [4] and Kim et al. [5] accelerated the T2w reconstruction by incorporating the T1w image as a reference. Various frameworks, such as progressive neural networks, generative adversarial networks (GANs), and dilated convolutions [8], have been modeled for multimodal super-resolution (SR) reconstruction. It is noteworthy that multimodal reconstruction goes beyond MRI and has been effectively utilized in other applications, such as low-dose PET images, combining with other MRI scans to mitigate radiation exposure risks associated with PET/MRI [29].

Despite successful reconstruction on multiple sources, these studies still have not adequately addressed the significance of spatial alignment between the target and reference MRIs.

C. Multimodal MR Alignment

The precise and durable similarity measurements are very crucial, particularly in the context of multimodal or multi-contrast registration. However, due to substantial variations in appearance across different modalities, it challenges more for them in computer vision. Mutual information (MI) has demonstrated its effectiveness in various tasks of multimodal image registration, including PET-CT and PET-MRI. Additionally, the feature-based quantity, such as modality-independent neighborhood descriptor (MIND), utilized high-order appearance features to establish spatial correspondences [31]. Fan et al. [39] introduced GAN to measure similarity, where the paired images are required [14]. In contrast, Cao et al. [47] employed cross-modal synthesis to evaluate similarity, thereby simplifying the complex computation with a monomodal metric. To overcome potential spatial distortion caused by

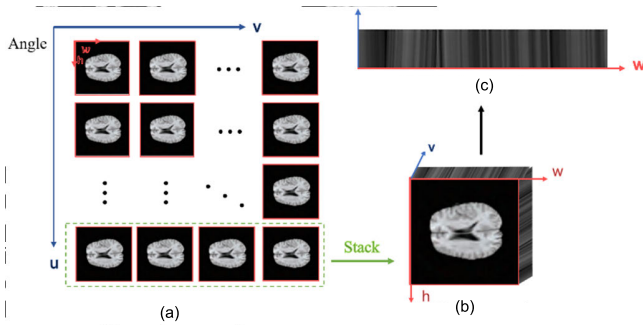


Fig. 1. (a) SAI, (b) 3-D MRI, and (c) EPI.

cross-modal synthesis, Arar et al. [48] presented a method to preserve geometric accuracy [39].

Above methods developed various similarity measurements to align spatial features from cross-modal MRIs, which are often fully sampled. However, for those undersampled MRIs, how to align the spatial features still needs further demonstration.

III. METHOD

A. Preliminary

An MRI can be conceptualized as a 4-D tensor as $L(u, v, h, w) \in R^{U \times V \times H \times W}$. In the representation, the variables U and V correspond to the angular dimensions of MRI data, while H and W represent the spatial dimensions. Due to the inherent complexity and the high dimensionality of 4-D tensors, direct visualization on them is infeasible. Consequently, MRI data are often structured and presented in a variety of formats to facilitate their interpretation and analysis.

1) *Subaperture Image (SAI)*: As illustrated in Fig. 1(a), an MRI can be organized as a (u, v) array with dimensions $H \times W$, which represents images obtained from different angular planes. Represented as $L(u, v, :, :) \in R^{H \times W}$ at the angular coordinates (u, v) , an SAI resembles a 2-D natural image, which aids in spatial information extraction.

2) *Epipolar Plane Image (EPI)*: As illustrated in Fig. 1(b) and (c), when those SAIs are along the angular dimension V , an EPI $L(u, :, h, :) \in R^{V \times W}$ is generated by stacking the 3-D images on the spatial angular plane. Given that objects in a space can be projected onto various spatial locations across different angles, an EPI displays oriented line patterns, with the slopes reflecting disparity values. Consequently, EPIs are widely employed in the inference of 3-D scene geometries for MRI reconstruction or depth estimation. Nevertheless, since an EPI only represents a 2-D horizontal or vertical slice of an MRI, it poses challenges in encoding the spatial and angular information.

3) *Macropixel Image (MacPI)*: Similar to SAI arrays, an MRI can be structured as an $H \times W$ arrangement of macropixels. A macropixel $L(:, :, h, w) \in R^{U \times V}$ is characterized as a collection of pixels that share identical spatial coordinates (h, w) , which are captured by cameras positioned at varied angular orientations. As depicted in Fig. 2, the transformation of an SAI array into an MacPI necessitates the pixels from each SAI at common spatial locations. Following

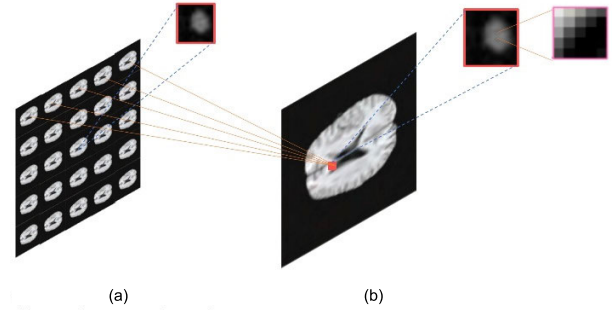


Fig. 2. Illustration of the connection between (a) SAI and (b) MacPI by transforming SAI array into an MacPI. To achieve this, pixels positioned identically in each SAI must be reorganized according to their angular coordinates to form a macropixel, as demonstrated by the 5×5 patch shown in the top-right corner of (b) resulting macropixels need to be sorted according to their spatial coordinates.

this, it is essential to organize the resulting macropixels based on their spatial coordinates. It is important to observe that the “mosaic pattern” produced by macropixels [illustrated in the magnified section of Fig. 2(b)] might not be favorable for human visual perception. However, MacPI enables a balanced integration of spatial and angular data within an MRI, thereby supporting the separation and assimilation of the spatial and angular features.

B. Motivation

In general, there are much more correlations in cross-modal MRIs, which are crucial for MRI reconstruction. Therefore, to improve its performance, it can generate the HR T2w images using the information learned from MRIs with other modalities, that is, the LR T2w and HR T1w images. The original MRI is often represented as feature tensor with the hybrid spatial, angular, and EPI information. Each has shown its contribution on reconstructing MRI. Thus, some modules should be developed to spatially decouple them. In addition, MRI can be divided into the high- and low-frequency components, each of which plays different roles in generating SR MRI. For example, when reconstructing the HR T2w based on the LR T2w, according to MRA theory, the HR T1w can also be incorporated with its high-frequency component while suppressing the low-frequency part. It incurs another problem that how to separate the high-frequency from the low-frequency using deep modules. On the other hand, feature-fusing techniques should be also seriously considered, including integrating the high-frequency component of the HR T1w into the LR T2w, or incorporating the downsampled with the decoupled features.

Consequently, MFSD has been constructed, where both the spatial decoupling and filter modules should be developed to depart the spatial from the angular information, and to extract the high-frequency parts from the HR T1w, respectively. Meanwhile, to fully utilize the correlations between cross-modal images, other structures are also built to fuse those features with residual strategy for further MRI reconstruction.

C. MRA Theory-Induced SR Reconstruction

MRA was initially introduced by Mallat [38]. MRA, fundamental to wavelet analysis, is widely utilized in various areas

of computer vision. Using MRA, the orthogonal decomposition to a signal space $L2(R)$ can be articulated at the following manner:

$$V_0 = V_1 \oplus W_1 = V_2 \oplus W_2 \oplus W_1 = \dots = V_N \oplus (\oplus_{k=1}^N W_k) \quad (1)$$

where W_1 represents the orthogonal component space of V_1 contained in V_0 . Space V_0 can be represented by a finite set of subspaces. For a signal $f(x, y) \in V_0$, it is decomposed into an approximate component V_N and a detailed component W_N . As a substitution, above decomposition to a signal $f^0(x, y)$ will be represented as

$$f^0(x, y) = f^N(x, y) + \sum_{k=1}^N d^k(x, y). \quad (2)$$

Thus, we employ signal decomposition to differentiate the $f^N(x, y)$ from the detail parts $d^k(x, y)$ ($k = 1, \dots, N$), allowing constructing the HR MRI from the LR MRI by merging the high-frequency details from the associated approximate images. For example, the T1w and the T2w are different images of a subject, showing notable relationship and likeness in their boundary shapes. According to MRA, it can capture the high-frequency features as prior information from the HR T1w to reconstruct the HR T2w, which is stated below

$$\begin{aligned} f^0(\text{T2w}) &= f^N(\text{T2w}) + \sum_{k=1}^N d^k(\text{T2w}) \\ &\approx f^N(\text{T2w}) \\ &\quad + \sum_{k=1}^N d^k(\text{T1w}) \end{aligned} \quad (3)$$

where $f^0(\text{T2w})$ represents the generated HR T2w image, $f^N(\text{T2w})$ denotes the LR T2w. Additionally, $d^k(\text{T1w})$ signifies the detail part derived from the HR T1w.

To this end, MFSD is constructed for indirect VBM.

D. MFSD Network

An overview of MFSD network is illustrated in Fig. 3. Starting with an SAI array $I_{\text{SAIs}}^{\text{LR}} \in R^{\text{AH} \times \text{AW}}$, MFSD converts it into an MacPI $I_{\text{MacPI}}^{\text{LR}} \in R^{\text{AH} \times \text{AW}}$ and then applies the spatial, angular, and EPI convolutions to decouple features in MacPI pattern. Inspired by residual channel attention networks (RCANs) [27], MFSD also introduces a residual-in-residual structure composed of two residual groups [42], [43], to fuse features from the cross-modal MRIs at different scales, improving the reconstruction performance. As shown in Fig. 3, three SDB-Groups, dense block (DB)#1, LFIF#1, and DB#2 constitute the first residual structure. By absorbing it, DB#3, LFIF#2, and DB#4 shape another residual-in-residual framework. The final output of MFSD is also the SAI array $I_{\text{SAIs}}^{\text{LR}} \in R^{\alpha \text{AH} \times \alpha \text{AW}}$, where α represents an upscaling factor. Namely, the LR T2w is taken as input by MFSD to generate the HR T2w as result of VBM.

MFSD contains four units: spatial feature extractor (SFE), spatial decoupled block (SDB), LFIF, and DB.

1) *Spatial Feature Extractor*: SFE extracts spatial features from MacPI, which is essential to convolve pixels from the same view while isolating those from different views of MacPI. Therefore, SFE is in practical a convolutional layer with a 3×3 kernel, a stride of 1, and a dilation factor of A . As a result, only the spatial information in MRI can be extracted. In addition, zero padding is introduced to ensure that its output maintains the same spatial dimensions as MacPI.

SFE captures the spatial features from the input as

$$F_I = H_{\text{SFE}}(I) \quad (4)$$

where $H_{\text{SFE}}()$ denotes a convolutional transformation, and I represents the input image, while F_I denotes the spatial features obtained through the convolution.

2) *Spatial Decoupled Block*: SDB is taken to derive domain-specific features from MacPI by separating MRIs. As the fundamental component of MFSD, illustrated in Fig. 3(b), SDB receives MacPI features $F_{\text{in}} \in R^{\text{AH} \times \text{AW} \times C}$ as input, where C , denoting the channel count, is set to 64. This framework consists of four parallel branches to disentangle features: one dedicated to angles, another for spatial data, and two for EPIs. Notably, in the spatial branch, two SFEs operate sequentially to amalgamate spatial information from each view. It is important to highlight that the channel count in this branch matches that of the input features. Consequently, the spatial branch produces the output $F_{\text{spa}} \in R^{\text{AH} \times \text{AW} \times C}$.

In the angular branch, an AFE produces the intermediate angular feature, denoted as $F_{\text{ang}}^{\text{tmp}} \in R^{H \times W \times (C/4)}$. Subsequently, it is upsampled by a factor of A as the output $F_{\text{ang}} \in R^{\text{AH} \times \text{AW} \times (C/4)}$. Due to the uneven distribution of pixels in a MacPI caused by edges and occlusions in real scenes [4], a 1×1 convolution along with a 2-D pixel shuffling layer (pixSF-2D), illustrated in Fig. 3(e), is adopted for angular-to-spatial upsampling.

Two EPI branches, including horizontal EPI (EPI-H) and vertical EPI (EPI-V), are adopted as feature extractors to partition the 4-D structure (4DS) into vertical-width (V-W) and angle-height (U-H) subdomains, respectively. EPI-H takes a 1×1 convolution layer for 1-D pixel shuffling (pixSF-1D), extracting features from the V-W subdomain. Similar to EPI-H, EPI-W also inserts a pixSF-1D layer to learn features from the U-H subdomain.

Both EPI-H and EPI-W utilize the pixSF layer. By rearranging group values along the channel dimension into pixel blocks along either height or width [pixSF-1D in Fig. 3(d)] or even both [pixSF-2D in Fig. 3(e)], it upsamples any image with an arbitrary factor, that is, multiplying these dimensions by a constant, thereby mitigating the checkerboard artifacts commonly associated with deconvolutional up-sampling. In the task of $\times r$ (where $r = 2, 3, 4$), features with size of $\text{AH} \times \text{AW} \times C_{\text{in}}$ are input into a convolutional channel, that is, a pixSF layer, as a result, they are shuffled to $\text{AH} \times \text{AW} \times r^2 C_{\text{out}}$.

In SDB, SFE further works on the features from four branches, whose output is then fused to the original input of SDB using the residual strategy.

3) *Low-Frequency Information Filtering*: LFIF contributes to extracting the high-frequency parts while filtering the low-frequency ones from the HR T1w images, whose structure

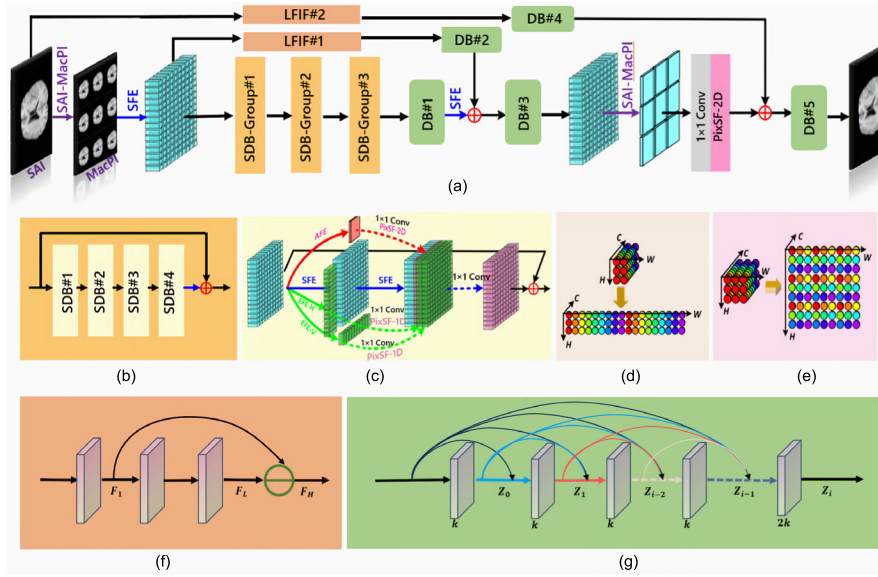


Fig. 3. (a) Overall architecture of MFSD, (b) spatial decoupled group, (c) SDB, (d) 1-D pix shuffling (PixSF-1D), (e) 2-D pix shuffling (PixSF-2D), (f) LFIF, and (g) DB.

can be found in Fig. 3(f). Thus, LFIF enhances the resolution of the T2w images by merging the LR T2w with the high-frequency details of the HR T1w, whose low-frequency information is removed. The process starts with downsampling the high-frequency features from the HR T1w images and follows an upsampling. Subsequently, the upsampled features are integrated with the LR T2w. Denoting F_I as the output of SFE and F_L as the features after upsampling, thus, we have

$$F_H = F_I - F_L \quad (5)$$

where F_H is the high-frequency feature.

4) *Dense Block*: DB with dense connections facilitates network training and mitigates gradient disappearance [31]. DB is introduced in MFSD as an alternative to batch normalization (BN) layers. Although BN layers are commonly used in original densely connected networks, they can impede network training. Consequently, BN layer is omitted. Additionally, inspired by MemNet [32], we introduce an $1 \times 1 \times 1$ convolutional layer at the end of DB to flexibly regulate the output, fulfilling the dual roles of feature fusion and dimensionality reduction. Within DB, the input for a given layer is derived from the concatenated outputs of all prior layers, as depicted in Fig. 3(g). Therefore, the output of DB can be represented as follows:

$$Z_i = H_{DB}([Z_0, Z_1, Z_2 \cdots Z_{i-1}]) \quad (6)$$

where $[Z_0, Z_1, Z_2 \cdots Z_{i-1}]$ denote the output from the 0th to the $(L-1)$ th convolutional layers, $H_{DB}()$ indicates a $1 \times 1 \times 1$ convolutional layer.

E. Spatially Decoupling

Spatially decoupling is suggested to transform the original MacPI into the spatial, angular, and other features in different 2-D subspaces using SFE, AFE, and EPI (EPI-H and EPI-W), displayed in Fig. 4.

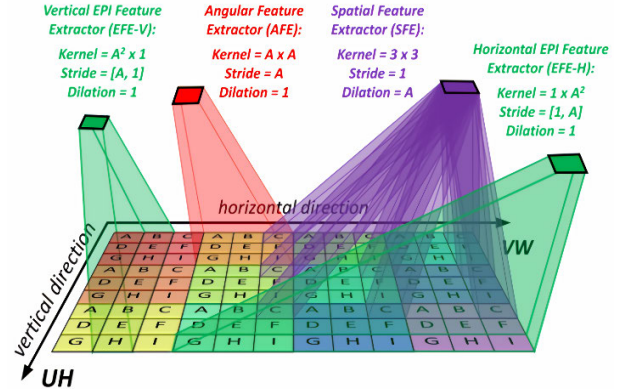


Fig. 4. Illustrations of the spatial, angular, and EPI feature extraction. In this scenario, an MRI model with dimensions $U = V = 3$ ($A = 3$), $H = 3$, and $W = 4$ is employed as an example. To represent MacPI, various macropixels are depicted against unique backgrounds, while pixels from different perspectives are shown using distinct characters. The introduced feature extractors are designed to segregate MRIs into various subspaces: SFE processes pixels from identical views, AFE works on pixels from the same macropixel group, and EFE deals with pixels on EPIs.

In these feature extractors, we set $U = V = A$, where A denotes the angular resolution. Since SFE has been described in advance, AFE and EPI will be discussed in the following.

1) *Angular Feature Extractor*: It is important to emphasize that only pixels residing within a micropixel undergo convolution, whereas pixels located in different macropixels remain distinct and cannot be aliased. An AFE has been introduced to extract angular features from MacPI. For the purpose, a convolutional layer has been designed with an $A \times A$ kernel and a stride of A . No padding is utilized, leading to an output with spatial dimensions of $H \times W$. The $A \times A$ kernel enables the convolution process for pixels contained within a macropixel, and the stride of A ensures that pixels from different macropixels are not aliased. Thus, this method can effectively learn the angular information of MRI.

2) *Epipolar Plane Image*: While SFE and AFE separate MRIs into the spatial and angular subspaces, their

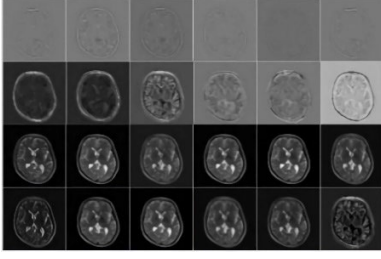


Fig. 5. Features obtained by various modules at the initial scale as LFIF#1 (row1), DB#1 (row2), DB#2 (row3), and the fusion module (row4).

interconnections are frequently neglected. Consequently, MFSD presents the horizontal and vertical EPs (EFE-H and EFE-V) to divide MRI into $V - W$ and $U - H$ subspaces, respectively. In contrast to SFE and AFE, EFE-H and EFE-V adopt asymmetric kernels and strides. As shown in Fig. 4, to extract features from the $V - W$ subspace, the pixels at a horizontal slice (where U and H coordinates are the same) of MacPI must undergo convolution. For this purpose, EFE-H, a convolution with a kernel of $1 \times A^2$, a vertical stride of 1, and a horizontal stride of A , is implemented without padding, which ensures that the output maintains the spatial dimensions of $AH \times W$. As a result, pixels across each horizontal slice are convolved, preserving the structure of the MRI. Hence, EFE-H processes information from one spatial dimension (W) and one angular dimension (V) simultaneously, thus augmenting its ability to model spatial-angular correlations. A similar technique is applied to the pixels across each vertical slice when employing EFE-V.

F. MS Future Fusion

MFSD reconstructs the HR T2w images by fusing the detailed information of the HR T1w into the LR T2w at two different scales. At first, features learned by three SDB-groups and DB#1 are fused with those via LFIF#1 and DB#2 using the residual technique. As mentioned above, the high-frequency details captured by LFIF#1 from the HR T1w image, are often incomplete and indistinct, which can be significantly enhanced using DB#2 to filter out noise and accentuate them in the HR T1w image. Meanwhile, DB#1 generates features from the LR T2 image, characterized by blurred edges. By fusing features processed by DB#1 and DB#2, its edge contours are sharpened, preserving the precise details from the HR T1w, which can be found from Fig. 5.

The second feature fusion incorporates LFIF#2, DB#4 into an upsampling module, which are illustrated in Fig. 6. At this level, the integration significantly enhances its resolution. LFIF#2 and DB#4 are essential to improve the overall representation of the feature. The upsampling further increases the spatial resolution of the output, critical for achieving a detailed and accurate representation of the underlying structures. Thus, these components work synergistically to fuse these features, resulting in precise and informative outputs.

G. SR Reconstruction Loss

SR reconstruction generates the HR images from the LR counterparts. In practical scenarios, HR MRI scans frequently

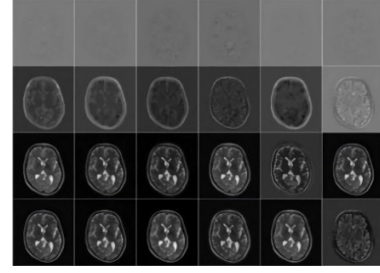


Fig. 6. Features obtained by various modules at the second scale as LFIF#2 (row1), DB#4 (row2), the up-sampling module (row3), and the fusion module (row4).

encounter issues, including downsampling, motion blur, and noise, during the acquisition and transmission phases. These challenges result in diminished resolution, which can be expressed as a linear equation

$$Y = HX + N = DBX + N \quad (7)$$

where X and Y denote the HR MRI and the LR MRI, respectively. In addition, D means downsampling, B represents blurring, and N is noise. Fusing D and B , it leads to matrix H .

SR reconstruction produces an HR image X with features from an LR image Y , which can be achieved using the least square error

$$\hat{X} = \underset{X}{\operatorname{argmin}} \|Y - HX\|^2 \quad (8)$$

where $\|\cdot\|$ denotes $L2$ norm. Equation (8) easily incurs unstable solutions, often mitigated by incorporating a regularization term $R(X)$ associated with prior knowledge

$$\hat{X} = \arg \left\{ \min_X \|Y - HX\|^2 + \lambda R(X) \right\} \quad (9)$$

where λ is a trade-off parameter.

However, selecting the regularization term is very challenging. It influences the outcomes such as no consistently reconstructing the HR image accurately. The parameter λ balances the regularization term and the prior information. One method can implicitly avoid the regularization term by training a model with an end-to-end mapping. Thus, it can be rewritten to

$$\hat{X} = \arg \min_X \|X - H^{-1}Y\|^2 \quad (10)$$

where H^{-1} represents the recovery operator.

The operator F is used to express H^{-1} . In particular, when considering a set of K samples from the HR MRI $X(i)$ associated with the LR MRI $Y(i)$, the recovery operator F is estimated in the following way:

$$\hat{F} = \arg \min_F \sum_i^K \|X^{(i)} - F(Y^{(i)})\|^2 \quad (11)$$

where $\hat{F}$ denotes the mapping from the LR MRI to the HR MRI.

This model can be realized using a neural network with end-to-end learning trick trained by a considerable quantity of data samples. Information from the HR T1w image is logically

leveraged as prior knowledge to improve SR reconstruction performance from the LR T2w image. Thus,

$$\hat{F} = \arg \min_F \sum_i^K \|X_{\text{HRT2 } w}^{(i)} - F(X_{\text{HRT1 } w}^{(i)}, Y_{\text{LRT2 } w}^{(i)})\|^2 \quad (12)$$

$$\hat{L}_{\text{HR}} = \sum_i^K \hat{F}(X_{\text{HRT1 } w}^{(i)}, Y_{\text{LRT2 } w}^{(i)}) \quad (13)$$

where F is also introduced as the mapping from the LR T2w incorporating the HR T1w, to the HR T2w.

Therefore, the loss of MFSD is defined as follows:

$$\mathcal{L}_{\text{feat}} = \left\| \sum_{l=1,3,5} \lambda_{\text{feat}}^{(l)} F^{(l)}(X_{\text{HRT1 } w}^{(i)}, Y_{\text{LRT2 } w}^{(i)}) - L_{\text{HR}} \right\|_1 \quad (14)$$

where $F^{(l)}(l = 1, 3, 5)$ denotes the feature mapping of the l th layer in DB, and $\lambda_{\text{feat}}^{(1)} = 0.2, \lambda_{\text{feat}}^{(3)} = 0.2, \lambda_{\text{feat}}^{(5)} = 0.1$ are hyperparameters for the loss. L_{HR} denotes the ground truth HR MRI and $\hat{L}_{\text{HR}}$ is obtained using (13). Due to parameter sharing and local receptive field mechanism, each subset of units in adjacent feature map is identical. Specifically, the feature unit of the l th layer is solely dependent on the subset of units adjacent to the feature map of the $(l - 1)$ th layer. To reduce feature redundancy, DB module consists of three layers.

To prevent the occurrence of multiple MRI patches in spatially decoupled networks assigned to the same feature vector, the proposed loss function incorporates a pixelwise term $\mathcal{L}_{\text{pix}}$. It utilizes mean absolute error (MAE) to measure the disparity between $\hat{L}_{\text{HR}}$ and L_{HR}

$$\mathcal{L}_{\text{pix}}(\hat{L}_{\text{HR}}, L_{\text{HR}}) = \|\hat{L}_{\text{HR}} - L_{\text{HR}}\|_1. \quad (15)$$

Thus, the total loss $\mathcal{L}_{\text{MFSD}}$ is defined as

$$\mathcal{L}_{\text{MFSD}} = \mathcal{L}_{\text{pix}} + \lambda \mathcal{L}_{\text{feat}} \quad (16)$$

where λ is a trade-off parameter.

IV. EXPERIMENT

Datasets: Data from the National Alliance for Medical Image Computing (NAMIC) was utilized, containing multimodal MRIs by a 3T GE scanner. The NAMIC includes 20 brain MRI cases, each composed of both the T1w and T2w image data subsets. In the experiment, referring [43], we selected 14 samples with 42 000 images as training set, while the remaining five samples of 15 000 images as test. In addition, MRIs in All tech Medical Systems Company LTD were also adopted without any preprocessing, where the true values of measurand (the ground truth HR images) are unavailable [30]. Unfortunately, some quantity values cannot be calculated for them. Thus, they are just suitable for MRI reconstruction visualization.

Settings: Different influence quantity values were also considered. The upscale factors, that is, $\times 2, \times 3, \times 4$, were used. A total of 300 training rounds were established, implementing a dynamic strategy to adjust the learning rate. In particular, the learning rate was initially set to 1×10^{-4} for the first 100 rounds, subsequently decreased to 1×10^{-5} . The network

TABLE I
QUANTITY VALUES OF ABLATION STUDY WITH FACTOR $\times 2$

	PSNR-T1/dB		SSIM-T1		PSNR-T2/dB		SSIM-T2	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
T1Net	40.4184	0.61	0.9866	0.0279	—	—	—	—
T2Net	—	—	—	—	39.6782	1.41	0.9882	0.0350
NaNet	37.4902	0.97	0.9795	0.0069	37.5465	1.68	0.9827	0.0087
FullNet	42.7942	1.18	0.9904	0.0060	41.4732	1.24	0.9896	0.0085

was optimized with $\mathcal{L}_{\text{MFSD}}$ loss using the Adam optimization technique [33], where α was specified as 0.0001, β_1 as 0.9, and β_2 as 0.999. In addition, a batch size of 8 was employed. MFSD was run on a computer with two NVIDIA RTX 3060Ti GPUs.

Baselines: To evaluate the efficacy of MFSD in VBM, several comparison approaches were suggested, including bicubic interpolation, low-rank, and total variation regularizations (LRTV) [40], enhanced deep SR (EDSR) [31], very deep convolutional networks for SR (VDSR) [34], faster SR convolutional networks (FSRCNNs) [35], feedback adaptive weighted dense network (FAWDN) [41], the coarse-to-fine SR CNN (CFSRCNN) [36], and residual dense network (RDN) [37].

Quantities: To justify the results of MFSD, two quantities were introduced as various similarities between the true value of measurand (the ground truth MRI) and the result of measurement (the generated HR MRI), that is, peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). PSNR compares the degree of image distortion by noise, whose symbol is decibel. In contrast, SSIM is a quantity for scenarios where the perceived image is prioritized over numerical distortion, particularly in medical image generation. PSNR exhibits its limitations in measuring the quality of multimodal medical images and aligning fully with human vision. Thus, it facilitates a more comprehensive quality measurement employing SSIM to integrate information of various modalities.

Procedure: We adopted the following measurement procedure when performing MFSD. With the same repeatability conditions, including the input, network, influence quantity values (parameters), and instrument (computer), MFSD was repeated ten times, generating ten HR MRI images, respectively. Accordingly, the quantity values of PSNR and SSIM can be calculated by taking into account each of the generated HR MRI images and the ground truth one. As a result, there are ten values for every quantity.

Uncertainty: We took type A evaluation of standard uncertainty [46]. For each quantity, mean and the experimental standard deviation (S.D.) of its values were obtained, where the latter characterizes the dispersion of the measurement results.

A. Ablation Study

- 1) The ablation study was conducted to verify the role of each main component of the network. At first, two SR networks T1Net and T2Net, were utilized for SR reconstruction on single-modal images. Both have introduced a single branch to take single-modal MRI (either T1w or T2w) as input in advance. In addition, an NaNet

TABLE II

QUANTITY VALUES OF ABLATION STUDY WITH FACTOR $\times 3$

	PSNR-T1/dB		SSIM-T1		PSNR-T2/dB		SSIM-T2	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
T1Net	34.5834	1.47	0.9673	0.0153	—	—	—	—
T2Net	—	—	—	—	35.1879	1.56	0.9730	0.0152
NaNet	33.3236	1.24	0.9527	0.0031	33.3085	1.33	0.9443	0.0044
FullNet	37.4982	1.08	0.9785	0.0039	36.6943	1.21	0.9762	0.0070

TABLE III

QUANTITY VALUES OF ABLATION STUDY WITH FACTOR $\times 4$

	PSNR-T1/dB		SSIM-T1		PSNR-T2/dB		SSIM-T2	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
T1Net	32.8301	1.42	0.9521	0.0144	—	—	—	—
T2Net	—	—	—	—	32.6644	1.41	0.9531	0.0164
NaNet	31.1321	1.18	0.9236	0.0020	31.0668	1.53	0.9827	0.0044
FullNet	34.7700	1.09	0.9635	0.0024	34.0168	1.38	0.9589	0.0081

removing the fusion modules from MFSD and a FullNet retaining all the structures of MFSD, were employed to evaluate its performance.

The quantity values by varying magnification factors ($\times 2$, $\times 3$, $\times 4$) were reported in Tables I–III. From these tables, it is evident that all the means of PSNR and SSIM by FullNet are the best compared with those using the other networks with the same magnification factor, demonstrating the advantages of the cross-modal correlation information between images for SR reconstruction. In addition, FullNet outperforms NaNet because the former fully takes advantages of the disentangled features including the angular and the spatial while the latter does not. Thus, it further validates that the disentangling module can significantly improve MRI quality in SR reconstruction. However, its S.D. is lower than that using NaNet in a few cases.

- 2) To evaluate the contributions of LFIF and multichannel structures, we constructed a multiscale (MS) structure by removing LFIF#1 and LFIF#2 filters, while compressing the multichannel into a single-channel widening channel widening (CW) structure. In the experiment, we take LRTV as baseline. Ablation test was performed on all MRI primitive data (PD) at a SR scale factor of $\times 2$ when using CW, MS, or both.

The results were presented in Table IV and Fig. 7. In Table IV, for CW, “0” indicates no CW, while “1” signifies the widening of the output channel of the middle layer using different expansion scale factors of 2 or 4. It is evident that merging these approaches results in a marginal enhancement of SR. The CW and MS components significantly improve model performance, yielding an increase of almost 0.9 dB. Notably, CW and MS together produced the most substantial improvements of 1.3 dB or so. Furthermore, Table IV illustrates that the quantity values improve with the increasing of the number of input channels; for instance, the performance with 64 channels surpasses that of 32 channels. From Fig. 7, it can conclude that at the same epochs, PSNRs using either CW or MS, even both go beyond that of baseline. Although it is always on the top by combining

TABLE IV

QUANTITY VALUES OF ABLATION STUDY WITH DIFFERENT COMBINATIONS

CW	Scale Factor	Input-Middle-Output	PSNR/dB		SSIM	
			Mean	S.D.	Mean	S.D.
0	1	32-32-32	38.74	1.19	0.9783	0.0029
0	1	64-64-64	39.19	1.23	0.9854	0.0035
1	2	32-64-32	40.28	1.20	0.9883	0.0038
1	2	64-128-64	41.53	1.18	0.9895	0.0037
1	4	32-128-32	41.59	1.19	0.9901	0.0026
1	4	64-256-64	41.63	1.09	0.9902	0.0024

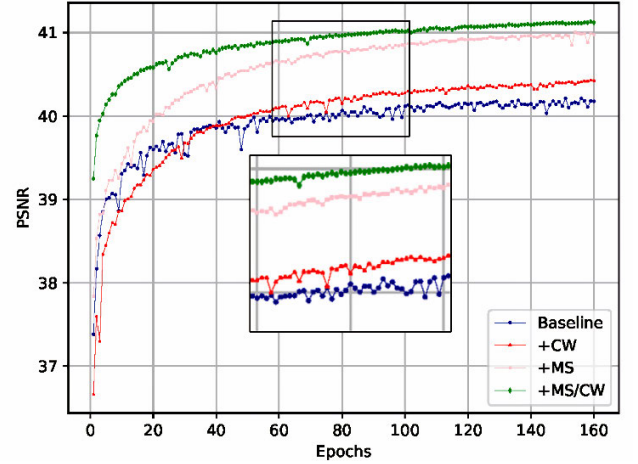


Fig. 7. PSNR curves by utilizing either one or both MS and CW on network, and baseline for SR $\times 2$.

CW with MS, the difference between it and the single MS is smaller and smaller.

B. Results

At first, we tested the impacts of network depth and spatial position on MRI reconstruction performance, then the results, including the quantity values, visualization, and the running time, were also reported in the following.

1) *Network Depth*: We examined the model’s performance by varying parameter t (the number of SDB), determining the overall depth of MFSD. Its depth is calculated as

$$D = 5t + u + 5 \quad (17)$$

where u is the depth for the pixSF layer based on the upscaling factor [13].

From Fig. 8, it easily observes that the quantity value of PSNR increases with epochs for any fixed t , no matter how many it is set to, that is, 3, 6, 9, or 12. Moreover, when t equals to 12, it achieves the largest quantity value at the same epoch. Namely, more layers in MFSD, better result of SR.

2) *Spatial Position*: Originally utilized for image classification, class activation map (CAM) demonstrates a weighted linear combination of visual patterns located in different spatial positions, helping to identify the key areas of MRI for reconstruction. We examined the CAM diagrams before and after MFSD, as shown in Fig. 9(a). Furthermore, we also presented a detailed comparison of local LFIF in Fig. 9(b). Fig. 9(a) shows that MFSD can improve the quality of the reconstructed image using CAM by making a comparison with the original

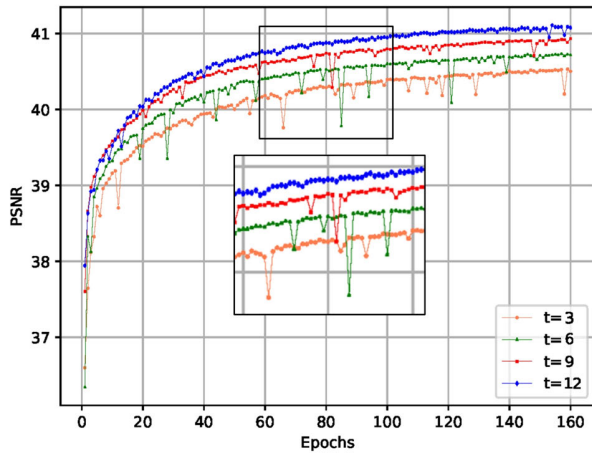


Fig. 8. PSNR curves with different network depth.

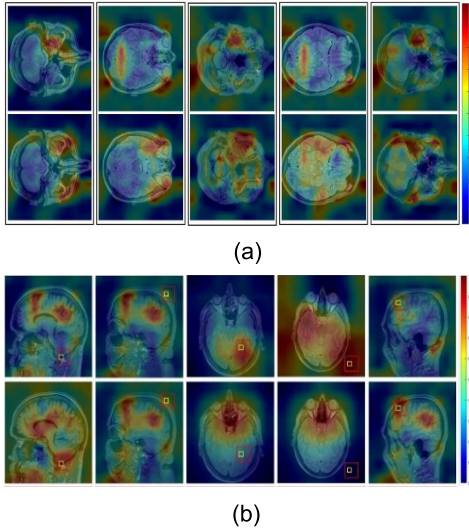


Fig. 9. Visualization of CAM before [the first row of (a) or (b)] and after [the second row of (a) or (b)] MFSD or LFIF. They reveal the significant discriminative regions. These regions are highlighted with red color indicating a high level of significance, while orange, green, and blue represent a gradual decrease in importance.

MRI, which is taken as the input of MFSD. In Fig. 9(b), CAM effectively displays the levels of saliency in different regions through a color-coded visualization. Regions highlighted in red indicate highly significant spatial positions, while those in orange, green, and blue represent spatial locations of minor importance.

3) *Quantity Values*: All comparison models, originally designed for natural images, were taken. By varying magnification factors ($\times 2$, $\times 3$, $\times 4$), the quantity values using these methods are presented in Tables V–VII. Table V displays that the proposed method achieves the top PSNR for SR reconstructed on both T1w and T2w modality at a magnification factor of $\times 2$. They gain the improvements of 0.1098 and 0.4155 dB, respectively, compared to those of the suboptimal method RDN. SSIM for T1w SR matches that of RDN, both ranking the first. For T2w SR, PSNR, and SSIM of MFSD are the best. Additionally, Table VI reveals that at a magnification factor of $\times 3$, its PSNR and SSIM for T1w SR are slightly lower than RDN, with a few decreases of 0.0646 and 0.0003 dB, respectively, while PSNR and SSIM for T2w SR remain the top. Table VII further demonstrates

TABLE V
QUANTITY VALUES USING DIFFERENT METHODS WITH
UPSCALE FACTOR $\times 2$

Methods	PSNR-T1/dB		SSIM-T1		PSNR-T2/dB		SSIM-T2	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Bicubic	34.7751	2.44	0.9763	0.0055	33.9612	2.01	0.9714	0.0135
LRTV	32.1396	2.29	0.9706	0.0044	34.3628	2.10	0.9660	0.0109
EDSR	35.2451	2.33	0.9818	0.0069	36.3944	2.12	0.9811	0.0090
FSRCNN	41.2435	2.39	0.9889	0.0060	37.3074	2.21	0.9863	0.0065
VDSR	36.8457	2.40	0.9843	0.0053	36.6337	2.25	0.9829	0.0056
FAWDN	38.5073	2.44	0.9864	0.0034	37.9739	2.27	0.9845	0.0057
CFSRCNN	42.5594	2.41	0.9802	0.0034	40.7549	2.23	0.9889	0.0060
RDN	42.6844	2.26	0.9904	0.0037	41.0577	2.26	0.9892	0.0059
MFSD	42.7942	2.53	0.9904	0.0032	41.4732	2.33	0.9896	0.0053

TABLE VI
QUANTITY VALUES USING DIFFERENT METHODS WITH
UPSCALE FACTOR $\times 3$

Methods	PSNR-T1/dB		SSIM-T1		PSNR-T2/dB		SSIM-T2	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Bicubic	32.1866	0.66	0.9536	0.0020	31.3384	1.66	0.9442	0.0102
LRTV	33.0696	0.78	0.9577	0.0038	32.2007	1.78	0.9500	0.0044
EDSR	31.9819	1.19	0.9616	0.0067	33.1050	1.76	0.9590	0.0043
FSRCNN	36.2874	0.77	0.9741	0.0072	34.5343	1.43	0.9694	0.0124
VDSR	33.9688	1.18	0.9681	0.0034	33.5280	1.41	0.9644	0.0052
FAWDN	34.8628	1.23	0.9691	0.0038	34.0920	1.53	0.9638	0.0062
CFSRCNN	37.1919	1.20	0.9779	0.0037	36.1311	1.43	0.9742	0.0050
RDN	37.5628	1.09	0.9788	0.0037	36.3984	1.26	0.9756	0.0059
MFSD	37.4982	1.23	0.9785	0.0024	36.6943	1.38	0.9762	0.0081

TABLE VII
QUANTITY VALUES USING DIFFERENT METHODS WITH
UPSCALE FACTOR $\times 4$

Methods	PSNR-T1/dB		SSIM-T1		PSNR-T2/dB		SSIM-T2	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Bicubic	30.3506	1.03	0.9278	0.0044	29.5851	1.70	0.9143	0.0431
LRTV	30.7152	1.71	0.9349	0.0042	29.9546	1.63	0.9233	0.0094
EDSR	29.6456	1.87	0.9311	0.0068	30.6657	1.73	0.9283	0.0093
FSRCNN	33.7318	1.80	0.9555	0.0045	32.7460	1.86	0.9490	0.0099
VDSR	31.8128	0.92	0.9442	0.0042	31.4115	1.89	0.9382	0.0093
FAWDN	32.7359	1.82	0.9475	0.0038	31.8622	1.24	0.9381	0.0091
CFSRCNN	34.7271	1.82	0.9631	0.0043	33.5439	1.82	0.9565	0.0100
RDN	34.6987	1.16	0.9639	0.0033	33.7475	1.44	0.9580	0.0084
MFSD	34.7700	1.09	0.9635	0.0029	34.0168	1.20	0.9589	0.0075

that at a magnification factor of $\times 4$, the proposed method achieves the highest PSNR for either T2w or T1w SR, with improvements of 0.2693 and 0.0713 dB over the second ranked RDN, respectively. However, SSIM for T1w SR is lower than that of RDN, while for T2w SR it is still the optimal. Thus, we can conclude that MFSD has its superiority over the comparison methods across all magnification factors, including the traditional methods as Bicubic and LRTV, and deep learning ones.

4) *Visualization*: We also visualized MRI reconstruction using different comparison methods with factor of $\times 4$, shown in Fig. 10. To make comparisons, we took another data of two in vivo T1 images obtained from the Medical Systems Company LTD without any preprocessing [30], where their true values or ground truth HR images are unavailable. Therefore, to enhance the visual clarity, we cropped and enlarged the blue and red boxes in the original LR T1w images. From Fig. 10, it observes that SR reconstruction by MFSD shows clearer contour (subfigures in the up two rows) and texture (subfigures in the down two rows) than other approaches.

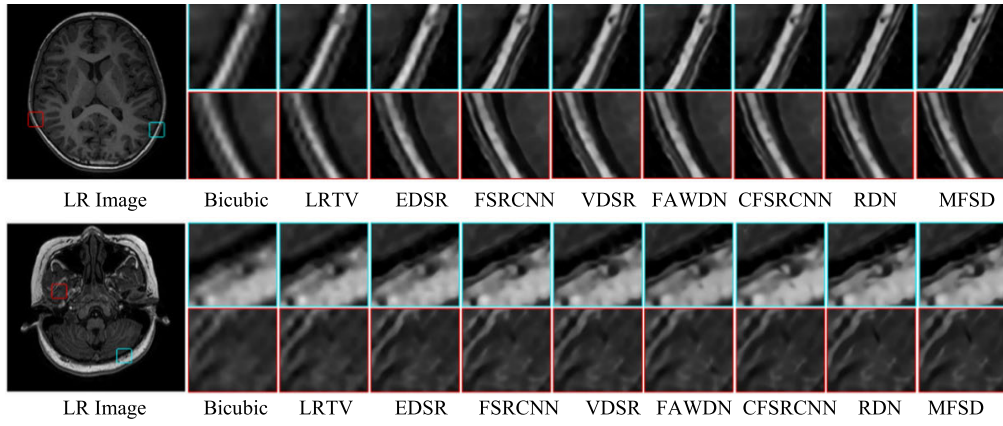


Fig. 10. Visualization of SR using different methods on in vivo T1 MRIs with SR $\times 4$.

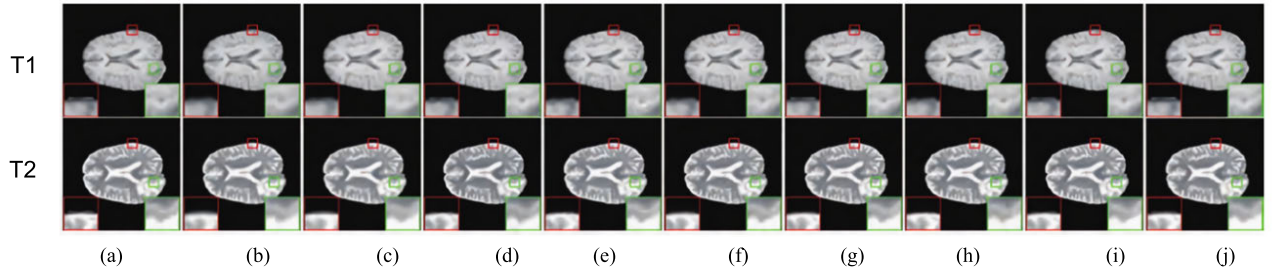


Fig. 11. Results using different methods with scale factor $\times 2$. (a) Ground truth. (b) Bicubic. (c) LRTV. (d) EDSR. (e) FSRCNN. (f) VDSR. (g) FAWDN. (h) CFSRCNN. (i) RDN. (j) MFSD.

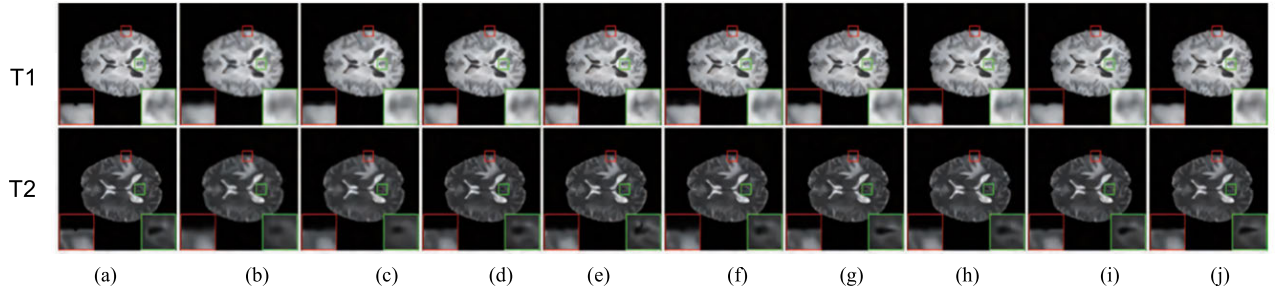


Fig. 12. Results using different methods with scale factor $\times 3$. (a) Ground truth. (b) Bicubic. (c) LRTV. (d) EDSR. (e) FSRCNN. (f) VDSR. (g) FAWDN. (h) CFSRCNN. (i) RDN. (j) MFSD.

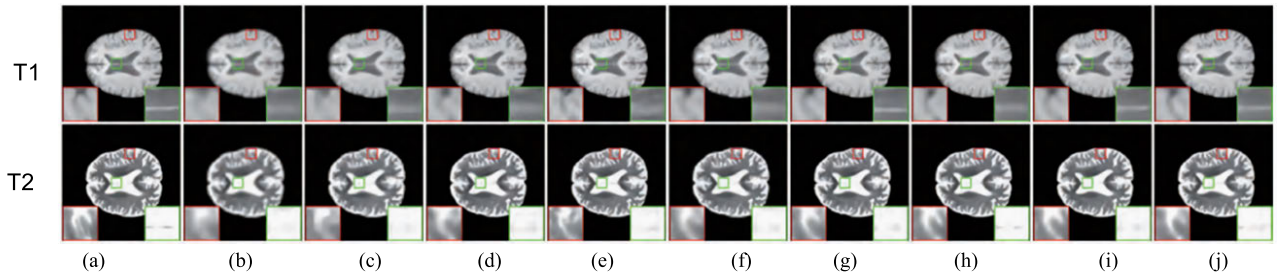


Fig. 13. Results using different methods with scale factor $\times 4$. (a) Ground truth. (b) Bicubic. (c) LRTV. (d) EDSR. (e) FSRCNN. (f) VDSR. (g) FAWDN. (h) CFSRCNN. (i) RDN. (j) MFSD.

In addition, we explored the relations between SR reconstruction and the magnification factors ($\times 2$, $\times 3$, $\times 4$) by using all the comparison algorithms, and the visualization results were displayed in Figs. 11–13, respectively. Each figure presents two local magnification regions for comparative analysis. These illustrations demonstrate that the proposed

MFSD outperforms the other approaches in preserving image details. Moreover, much more MRI details will be offered by MFSD with much larger magnification factors.

5) *Parameters and Running Time*: The parameters involved in the comparison methods have also been provided, mainly incurring the computational payments for their optimization.

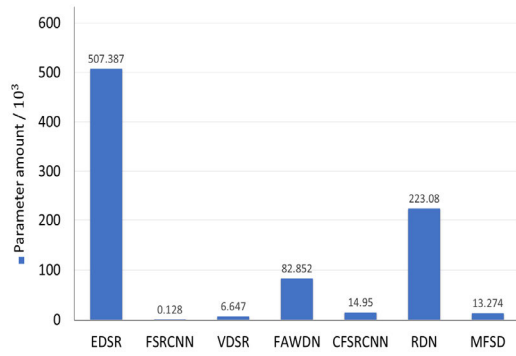


Fig. 14. Numbers of parameters of different methods.

TABLE VIII
QUANTITY VALUE OF AVERAGE RUNNING TIME BY DIFFERENT
METHODS FOR SR ON A SINGLE IMAGE

Methods	Time (ms)
EDSR	5.9
FSRCNN	1.5
VDSR	2.2
FAWDN	87.3
CFSRCNN	4.8
RDN	38.5
MFSD	2.3

Fig. 14 illustrates the parameter volume for each. Among them, FSRCNN has the smallest number of parameters due to its simple network structure; however, this simplicity results in lower quality HR images. VDSR also has a relatively small number of parameters, and its reconstruction quality is still inferior to that of MFSD. On the other hand, although the largest volume of parameters is involved in EDSR, it is still the minor to improve SR quality. RDN displays its SR reconstruction ability next to MFSD, while it necessitates approximately 17 $\times$ parameters.

Table VIII reports the quantity value of running time with symbol millisecond for MRI reconstruction, derived from the averaged results of 200 image sets. From Table VIII, it can find that FAWDN pays the most computational costs and FSRCNN requires the least expenses for SR. By fully considering Table VIII and Fig. 14, it also reveals a significant correlation between the network's runtime and the number of parameters. Although FAWDN has a relatively modest number of parameters, it exhibits the longest runtime for optimization through feedback techniques trained with a larger image dataset. In contrast, while RDN has fewer parameters than EDSR, its extensive use of dense connections also results in longer runtime. The methodology proposed in this study demonstrates the substantial computational efficiency, with a runtime approximately one-sixteenth that of RDN.

V. CONCLUSION

HR MRI is essential for accurate disease diagnosis due to its detailed anatomical information. However, challenges such as instrument limitations, prolonged scanning time, patient discomfort, and noise impede the measurement of MRI. What's more, much correlation information, crucial for SR, between cross-modal MRIs is often ignored, resulting in its low quality.

To address these issues, this article proposes a deep learning framework, termed MFSD, for image generation or indirect VBM from the LR T2w and the HR T1w. According to MRA theory, the network merges the spatially decoupled information from the LR T2w image with the high-frequency component of the HR T1w modality to generate the HR T2w image, enhancing its quality. Experiments on publicly available datasets reveal that MFSD is capable of SR or VBM by taking full advantage of the cross-modal features.

In MFSD, the proposed spatial decoupling enables the decomposition of 4-D tensor data into multiple subspaces, facilitating effective learning of the underlying structures of MRI. Quantity values in ablation test also prove that the spatially decoupled features contribute more to SR, demonstrating the effectiveness of our proposed method by utilizing the spatially decoupled features, integrating the detailed information of another modality.

Meanwhile, the proposed MFSD exhibits the improved performance on lightweight and generalization by making comparisons with some existing SOTA techniques. It is constructed with a streamlined architecture to reduce the volume of parameters, leading to comparatively fast running speed. The network generating high-quality MRIs across two different datasets, that is, the NAMIC and in vivo T1 MRIs, underscores its good generalization. Overall, MFSD is good at SR or indirect VBM, and its lightweight network makes it well suited for a variety of medical image SR applications.

While the proposed framework yields favorable results, there are still some limitations. For example, when fusing features for the further MRI reconstruction, MFSD currently presents a strategy to fuse them at two scales, confining SR task with a scaling factor $\times 3$. In future, we will develop a framework for VBM with a broader range of scaling factors.

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